

Interpretable Automated Ensemble Forecasting for Inflation: A Comprehensive Benchmarking Framework for Sri Lanka

Track: Digitalisation, AI, and Data in Macroeconomic Analysis

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ABSTRACT

Accurate inflation forecasting remains one of the most important analytical tasks, as inflation expectations directly influence monetary policy formulation, interest rate decisions, fiscal planning, financial market stability, and long-term economic growth. The challenge becomes particularly significant in emerging economies such as Sri Lanka, where inflation dynamics are influenced by structural breaks, exchange rate volatility, external commodity price shocks, supply-side disruptions, and policy regime changes. The unprecedented macroeconomic instability experienced during the recent economic crisis further highlighted the limitations of conventional statistical forecasting approaches that often struggle to adapt to rapidly evolving economic environments. While recent advances in machine learning and deep learning have demonstrated promising forecasting capabilities across various domains, their application to macroeconomic forecasting remains limited, particularly within the context of developing economies. Furthermore, there exists little empirical evidence regarding how modern automated forecasting frameworks combine multiple forecasting models, how these combinations evolve under different economic regimes, and whether such systems provide reliable and interpretable forecasts suitable for policy support.

This study proposes a comprehensive benchmarking and interpretability framework for inflation forecasting in Sri Lanka using a modern automated time-series forecasting framework that integrates statistical, machine learning, and deep learning forecasting models into a weighted ensemble. Rather than relying on a single forecasting algorithm, the framework automatically evaluates a diverse collection of forecasting models—including classical statistical methods, gradient boosting approaches, and state-of-the-art transformer-based deep learning models—and determines an optimal weighted ensemble according to the characteristics of the data and the selected evaluation criterion. The research seeks to answer four primary questions. First, how effectively can modern automated ensemble forecasting systems predict Sri Lankan inflation across different forecast horizons? Second, how do ensemble weights assigned to different forecasting models vary across inflation components, data frequencies, different evaluation metrics and economic regimes? Third, which forecasting evaluation metrics provide the most meaningful assessment of forecasting quality within the Sri Lankan macroeconomic context? Finally, how does the inclusion of macroeconomic explanatory variables and varying historical training lengths

influence forecasting accuracy and the relative contribution of recent deep learning foundation models?

The empirical analysis utilizes publicly available Sri Lankan macroeconomic data published by the World Bank. Four major inflation indicators are considered independently, namely the Headline Consumer Price Index (HCPI), Food Consumer Price Index (FCPI), Core Consumer Price Index (CCPI), and Energy Consumer Price Index (ECPI). To investigate the sensitivity of forecasting performance to data granularity, experiments are conducted using annual, quarterly, and monthly observations where applicable. This multi-frequency design enables an examination of whether different temporal resolutions alter the ensemble composition and forecasting behaviour of automated forecasting systems.

Forecasting performance is evaluated over a twelve-month prediction horizon using rolling-origin evaluation and multiple forecasting windows designed to represent distinct macroeconomic regimes, including relatively stable inflation periods and periods characterized by heightened volatility. Rather than reporting only overall forecasting accuracy, the study analyses how forecast errors evolve over a twelve-month prediction horizon while also evaluating quantile forecasts to assess the model's ability to characterize forecast uncertainty. This dual evaluation provides insights into both point prediction performance and the reliability of predictive intervals, which are particularly relevant for monetary authorities whose policy decisions depend on both expected inflation and the associated uncertainty across short- and medium-term horizons.

Recognizing that no single evaluation metric fully captures forecasting performance, the study develops a comprehensive evaluation framework encompassing multiple complementary dimensions of forecast quality. Deterministic forecasting performance is assessed using seven widely adopted point forecast accuracy measures that quantify prediction error from different statistical perspectives. Probabilistic forecasting capability is evaluated using proper scoring rules that measure both forecast calibration and uncertainty estimation, thereby examining whether predictive distributions accurately represent future inflation uncertainty. Beyond conventional error measures, temporal similarity between observed and predicted inflation trajectories is assessed using Dynamic Time Warping (DTW), Soft Dynamic Time Warping (Soft-DTW), Wasserstein Distance, and the Temporal Distortion Index (TDI). These metrics evaluate whether forecasting models successfully

preserve the underlying temporal evolution of inflation rather than merely minimizing pointwise prediction errors. Furthermore, directional forecasting capability is examined through Trend Direction Accuracy (TDA), Directional Accuracy (DA), and Turning Point Accuracy (TPA), providing insights into the ability of forecasting models to correctly anticipate inflation accelerations, decelerations, and major turning points that are particularly important for monetary policy decisions. Computational efficiency and execution time are also evaluated to assess the operational feasibility of deploying modern forecasting systems within institutional forecasting environments where timely forecast generation is essential.

An important contribution of this research lies in the interpretability of automated ensemble forecasting. Instead of treating automated forecasting systems as black-box predictors, the study systematically investigates how ensemble weights are allocated among constituent forecasting models under varying economic conditions. Particular attention is given to the evolving contribution of recent transformer-based foundation models for time-series forecasting relative to classical statistical models and conventional machine learning algorithms. By analysing these weight distributions across different inflation indices, data frequencies, evaluation metrics, and economic regimes, the study provides new empirical evidence regarding when advanced deep learning models contribute meaningful forecasting improvements and when simpler statistical approaches remain competitive. Such insights are expected to improve confidence in the adoption of modern forecasting systems.

Building upon the evaluation framework, the study further extends the forecasting models through multivariate forecasting experiments incorporating economically relevant explanatory variables. These include international crude oil prices, exchange rates, policy interest rates, global food prices, and selected domestic macroeconomic indicators such as broad money growth and industrial production, subject to data availability. These experiments investigate whether incorporating external and domestic macroeconomic information enhances predictive performance beyond univariate inflation forecasting and whether the relative importance of different forecasting models changes when richer economic information is introduced. Additional experiments systematically vary the historical training window to evaluate how the amount of available historical information influences forecasting performance and, in particular, the reliance placed upon recent deep learning foundation models within the automated ensemble.

The anticipated findings suggest that automated ensemble forecasting frameworks can deliver robust and competitive inflation forecasts across multiple inflation indices while adapting model selection dynamically according to changing economic conditions and forecast objectives. The study also expects to demonstrate that forecasting performance varies substantially depending on the evaluation metric employed, implying that model selection based solely on conventional point forecast errors may overlook important dimensions of forecasting quality relevant to policy applications. Furthermore, the analysis is expected to reveal that ensemble compositions evolve across economic regimes, forecast horizons, and training window lengths, providing valuable insights into the strengths and limitations of statistical, machine learning, and deep learning forecasting approaches under different macroeconomic environments.

The research contributes to both the macroeconomic forecasting and applied machine learning literature by providing one of the first comprehensive evaluations of automated ensemble time-series forecasting for Sri Lankan inflation. Beyond presenting forecasting accuracy comparisons, it establishes a practical framework for interpreting automated forecasting systems, selecting appropriate evaluation criteria, and understanding the circumstances under which modern deep learning models provide measurable benefits. To encourage transparency, reproducibility, and future methodological extensions, the complete forecasting framework, experimental configurations, and implementation code will be made openly available. The findings are expected to provide practical guidance for economists, central bank researchers, and policy analysts seeking to integrate modern artificial intelligence technologies into operational inflation forecasting while maintaining interpretability, robustness, and policy relevance. Ultimately, the proposed framework aims to support evidence-based forecasting practices that strengthen monetary policy analysis and improve macroeconomic decision-making in Sri Lanka and other emerging economies facing similar structural and economic challenges.